

Fusion tracking in color and infrared images using joint sparse representation

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Received March 29, 2011; accepted November 24, 2011

Abstract Currently sparse signal reconstruction gains considerable interest and is applied in many fields. In this paper, a similarity induced by joint sparse representation is designed to construct the likelihood function of particle filter tracker so that the color visual spectrum and thermal spectrum images can be fused for object tracking. The proposed fusion scheme performs joint sparse representation calculation on both modalities and the resultant tracking results are fused using min operation on the sparse representation coefficients. In addition, a co-learning approach is proposed to update the reference templates of both modality and enhance the tracking robustness. The proposed fusion scheme outperforms state-of-the-art approaches, and its effectiveness is verified using OTCBVS database.

Keywords joint sparse representation, fusion tracking, color image, infrared image

Citation Liu H P, Sun F C. Fusion tracking in color and infrared images using joint sparse representation. *Sci China Inf Sci*, 2012, 55: 590–599, doi: 10.1007/s11432-011-4536-9

1 Introduction

By leveraging the combined benefits of using different modalities while compensating for failures in individual modalities[1], multi-sensor systems offer considerable advantage over single-sensor systems. They benefit from both an increased spatial and temporal coverage due to complementary sensor ranges and data acquisition frequencies. In addition, they offer an improved operational robustness, because distinct physical sensing principles compensate for particular perception shortcomings. Altogether, they are able to gather more information about the reality especially in varying environmental conditions[2]. As indicated by [3], the rapid developments of sensor technology, microelectronics, and communications have led to a great need for image fusion techniques that can effectively combine multi-sensor images into an enhanced single view of a scene with extended information content. The use of color greatly expands the amount of information that can be conveyed in a single image, giving rise to a natural approach to the representation of multi-modal data. Refs. [4,5] also pointed out the importance of fusion of multiple sensor information.

For fusion tracking, the choice of visual and infrared imagery is significant, as each modal provides disparate, yet complementary information about a scene. Infrared cameras detect relative differences in

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the amount of thermal energy emitted from objects in the scene. These sensors are therefore independent of illumination, making them more effective than color cameras under poor lighting conditions. Color sensors on the other hand, are oblivious to temperature differences in the scene, and are typically more effective than thermal cameras when objects are at “thermal crossover”, provided that the scene is well illuminated and the objects have color signatures different from the background[6].

There are a lot of approaches for fusion of color and infrared images. To give a comprehensive review of this topic in this paper is impossible. So in this section only some related results are given. Ref. [7] presents a moving object detection and tracking system that robustly fuses infrared and visible video within a level set framework. The long-term trajectories for object clusters are estimated using Kalman filtering and watershed segmentation. Kalman filtering can obtain optimal solution in the case of linear dynamics and Gaussian noise. Unfortunately, very few practical visual tracking problems belong to this case. Using mean-shift approach, ref. [8] proposes a framework that can efficiently combine features for robust tracking based on fusing the outputs of multiple spatiogram trackers. However, mean-shift is a local deterministic search strategy, which is easy to be trapped into local minimal, and difficult to recover from tracking failure.

The particle filter, also known as sequential Monte Carlo[9], or Condensation[10], is the most popular nonlinear and non-Gaussian filtering approach. It recursively constructs the posterior probability distribution function of the state space using Monte Carlo integration[11]. One important advantage of the particle filtering framework is that it allows the information from different measurement sources to be fused in a principled manner[12]. Using particle filter to fuse the color and infrared images attracts many attentions. Ref. [1] evaluates the appearance tracking performance of multiple fusion schemes that combine information from color and thermal infrared images for the tracking of surveillance. In that paper, some common fusion algorithms are summarized and compared, including simple and weighted averaging, similarity score product, min and max score fusion and dynamic weighting approaches. Ref. [13] investigates the impact of pixel-level fusion of videos from visible and infrared surveillance cameras on object tracking performance, as compared to tracking in single modality videos. Tracking has been accomplished by means of a particle filter which fuses a color cue and the structural similarity measure. However, how to design a robust likelihood which effectively incorporates multiple modal information remains an open challenging problem.

Recently sparse signal reconstruction gains considerable interest[14]. Variations and extensions of l_1 minimization have been applied to many vision tasks[15]. In these works, using sparsity as a prior leads to excellent results. In [16], a robust visual tracking framework is developed by casting the tracking problem as finding a sparse approximation in a template subspace. During tracking, a target candidate is represented as a linear combination of the template set composed of both target templates and trivial templates. Intuitively, a good target candidate can be efficiently represented by the target templates. This leads to a sparse coefficient vector, since coefficients corresponding to trivial templates tend to be zeros. The sparse representation is achieved through solving an l_1 -regularized least squares problem. Then the candidate with the smallest target template projection error is chosen as the tracking result. In [17], a multi-part subspace appearance model for improving robustness is developed and an over-complete dictionary is learned off-line. However, particle filters are used in both [16] and [17] and the sparse representation needs to be calculated for each particle. Therefore the time cost is very large. To tackle this problem, ref. [18] gives a different way of treating tracking task as sparse approximation problem. The advantages of this approach is that the sparse representation needs to be calculated only once and the time cost is dramatically decreased. In addition, sparse representation provides a new approach to construct the likelihood function. However, this work is restricted to single sensor cases and the extension to multiple sensor fusion is nontrivial.

In this paper, the joint sparse presentation approach is utilized to design fusion tracking approach for color and infrared images. This approach shares the same calculation advantage of the approach proposed in [18] and provides better robustness. In addition, a co-learning-like approach is proposed to mutually update the reference templates of color and infrared images. This approach dramatically reduces the risk of drift which often occurs in self-learning updating approach.

The organization of this paper is as follows: Section 2 gives a brief review of particle filter in color and infrared images. The new likelihood function is introduced in detail in Section 3. Section 4 presents the co-learning updating approach. The experimental results are given in Section 5. Finally, Section 6 gives some conclusions.

2 Brief review of particle filter

The task of tracking is to use the available measurement information to estimate the hidden state variables. Given the available observations $\mathbf{z}_{1:k-1} = \mathbf{z}_1, \mathbf{z}_2, \dots, \mathbf{z}_{k-1}$ up to time instant $k-1$, the prediction stage utilizes the probabilistic system transition model $p(\mathbf{x}_k|\mathbf{x}_{k-1})$ to predict the posterior at time instant k as

$$p(\mathbf{x}_k|\mathbf{z}_{1:k-1}) = \int p(\mathbf{x}_k|\mathbf{x}_{k-1})p(\mathbf{x}_{k-1}|\mathbf{z}_{1:k-1})d\mathbf{x}_{k-1}. \quad (1)$$

At time instant k , the observation $\mathbf{z}_k$ is available, and the state can be updated using Bayes's rule

$$p(\mathbf{x}_k|\mathbf{z}_{1:k}) = p(\mathbf{z}_k|\mathbf{x}_k)p(\mathbf{x}_k|\mathbf{z}_{1:k-1})/p(\mathbf{z}_k|\mathbf{z}_{1:k-1}), \quad (2)$$

where $p(\mathbf{z}_k|\mathbf{x}_k)$ is described by the observation equation. The kernel of particle filter is to recursively approximate the posterior distribution using a finite set of weighted samples. Each sample $\mathbf{x}_k^i$ represents one hypothetical state of the object, with a corresponding discrete sampling probability ω_k^i , which satisfies $\sum_{i=1}^N \omega_k^i = 1$. The posterior $p(\mathbf{x}_k|\mathbf{z}_{1:k})$ then can be approximated as

$$p(\mathbf{x}_k|\mathbf{z}_{1:k}) \approx \sum_{i=1}^N \omega_k^i \delta(\mathbf{x}_k - \mathbf{x}_k^i), \quad (3)$$

where $\delta(\cdot)$ is Dirac function. Then the estimation of the state $\mathbf{x}_k$ can be obtained as $\hat{\mathbf{x}}_k = \sum_{i=1}^N \omega_k^i \mathbf{x}_k^i$. In standard particle filter the weight ω_k^i becomes proportional to the observation likelihood $p(\mathbf{z}_k|\mathbf{x}_k^i)$. For more details, please refer to [9]. In Figure 1 an illustration of the sampling procedure is given. It is obvious that the likelihood function $p(\mathbf{z}_k|\mathbf{x}_k)$ is very important for determining the goodness of each particle and needs careful designs. In the following the new design for this likelihood function will be introduced.

3 Proposed new likelihood function

Sparse induced similarity was first proposed in [19,20], this similarity to construct the graph-based semi-supervised learning. This similarity depends on two basic assumptions: linearity and sparsity. Linear assumption claims that any feature vector in a class can be represented as a linear combination of some other feature vectors in the same class. Sparsity assumption claims that given a feature vector, its sparsest representation is achieved when all the basis feature vectors belong to the same class as the feature vector.

For tracking in color images, one color reference template $\mathbf{r}_C \in \mathbb{R}^d$ can be obtained at the initial frame. Then, at time instant k , N particle $\{\hat{\mathbf{x}}_k^i\}_{i=1}^N$ will be generated. From these particles the candidate feature vectors $\{\mathbf{f}_k^i\}_{i=1}^N \in \mathbb{R}^{d \times N}$ can be extracted. In addition, a trivial template sets $\mathbf{I} = [\mathbf{I}_1, \mathbf{I}_2, \dots, \mathbf{I}_d] \in \mathbb{R}^{d \times d}$ can be defined, where the trivial template $\mathbf{I}_i \in \mathbb{R}^d$ is vector with only one nonzero entry. As a result, $\mathbf{I}$ is an identity matrix. Then $\mathbf{r}_C$ can be decomposed as a sparse linear combination of the feature vectors $\{\mathbf{f}_k^i\}_{i=1}^N$ and trivial templates $\{\mathbf{I}_i\}_{i=1}^d$, which can be combined into the following matrix:

$$\mathbf{F} = [\mathbf{f}_k^1 \ \mathbf{f}_k^2 \ \dots \ \mathbf{f}_k^N \ \mathbf{I}_1 \ \mathbf{I}_2 \ \dots \ \mathbf{I}_d] \in \mathbb{R}^{d \times (N+d)}. \quad (4)$$

Denote by $\boldsymbol{\alpha} = [\alpha_1 \ \alpha_2 \ \dots \ \alpha_{N+d}]^T$ the coefficients of the sparse decomposition. Then it can be defined as the solution of the following optimization problem:

$$\min_{\boldsymbol{\alpha}} \|\boldsymbol{\alpha}\|_0 \quad \text{s.t.} \quad \mathbf{F}\boldsymbol{\alpha} = \mathbf{r}_C, \quad (5)$$

where $\|\alpha\|_0$ denotes the 0 norm of α , i.e., the cardinal of the set of indices of nonzero components of α .

Similarly, for tracking in infrared images, a reference template $r_I \in \mathbb{R}^{\bar{d}}$ can be obtained at the initial frame. At time instant k , for the same N particle $\{\hat{x}_k^i\}_{i=1}^N$, the candidate feature vectors $\{\bar{f}_k^i\}_{i=1}^N \in \mathbb{R}^{\bar{d} \times N}$ can be extracted. In addition, the trivial template sets $\bar{I} = [\bar{I}_1, \bar{I}_2, \dots, \bar{I}_d] \in \mathbb{R}^{\bar{d} \times \bar{d}}$ can be defined, where the trivial template $\bar{I}_i \in \mathbb{R}^{\bar{d}}$ is a vector with only one nonzero entry. As a result, $\bar{I}$ is an identity matrix. Then r_I can be decomposed into a sparse linear combination of the feature vectors $\{\bar{f}_k^i\}_{i=1}^N$ and trivial templates $\{\bar{I}_i\}_{i=1}^d$, which can be combined into the following matrix:

$$\bar{F} = [\bar{f}_k^1 \ \bar{f}_k^2 \ \dots \ \bar{f}_k^N \ \bar{I}_1 \ \bar{I}_2 \ \dots \ \bar{I}_d] \in \mathbb{R}^{\bar{d} \times (N+\bar{d})}. \quad (6)$$

Denote $\bar{\alpha} = [\bar{\alpha}_1 \ \bar{\alpha}_2 \ \dots \ \bar{\alpha}_{N+\bar{d}}]^T$ as the coefficients of the sparse decomposition. Then it can be defined as the solution of the following optimization problem:

$$\min_{\bar{\alpha}} \|\bar{\alpha}\|_0 \quad \text{s.t.} \quad \bar{F}\bar{\alpha} = r_I. \quad (7)$$

Denote $A = [\alpha \ \bar{\alpha}]$, and $\|A\|_{2,0}$ to measure the number of rows in A that contain nonzero entries[21]. Then finding the jointly sparsest representation is equivalent to solving the following problem:

$$\min_A \|A\|_{2,0} \quad \text{s.t.} \quad F\alpha = r_C, \quad \bar{F}\bar{\alpha} = r_I. \quad (8)$$

However, solving the above problem requires enumerating all subsets of the set $\{1, 2, \dots, N\}$, which is essentially a combinatorial optimization problem and is in general NP-hard. Similar to the use of the 1-norm minimization to replace the 0-norm minimization, one natural alternative is to use $\|A\|_{2,1}$ instead of $\|A\|_{2,0}$, resulting in the following convex optimization problem (The relationship between the above two problem can be found in [22]):

$$\min_A \|A\|_{2,1} \quad \text{s.t.} \quad F\alpha = r_C, \quad \bar{F}\bar{\alpha} = r_I. \quad (9)$$

This problem can be further converted in the following form:

$$\min_{\alpha, \bar{\alpha}} \|\alpha\|_2 + \|\bar{\alpha}\|_2 + \lambda(\|\alpha\|_2 + \|\bar{\alpha}\|_2), \quad (10)$$

where λ is a regularization parameter and is usually set to 0.01. According to [23], the solving algorithm can be summarized in Algorithm 1.

Algorithm 1 Joint sparse representation

Input: $r_C, F, r_I, \bar{F}, \lambda, \eta$

Output: $\alpha, \bar{\alpha}$

- 1: Initialization: Properly initialize $\alpha^0, \bar{\alpha}^0, \beta^0, \bar{\beta}^0$. Set $\gamma_0 = 1$ and $t = 0$.
 - 2: **while** Not convergent **do**
 - 3: $\alpha^{t+1} = \beta^t - \eta[-F^T r_C + F^T F \beta^t]; \quad \bar{\alpha}^{t+1} = \bar{\beta}^t - \eta[-\bar{F}^T r_I + \bar{F}^T \bar{F} \bar{\beta}^t].$
 - 4:
 - 5: $\alpha^{t+1} = [1 - \frac{\lambda\eta}{\|\alpha^{t+1}\|_2}]_+ \alpha^{t+1}; \quad \bar{\alpha}^{t+1} = [1 - \frac{\lambda\eta}{\|\bar{\alpha}^{t+1}\|_2}]_+ \bar{\alpha}^{t+1}.$
 - 6:
 - 7: $\gamma^{t+1} = \frac{2}{t+3}.$
 - 8: $\beta^{t+1} = \alpha^{t+1} + \frac{(1-\alpha^t)\alpha^{t+1}}{\alpha^t}(\alpha^{t+1} - \alpha^t); \quad \bar{\beta}^{t+1} = \bar{\alpha}^{t+1} + \frac{(1-\bar{\alpha}^t)\bar{\alpha}^{t+1}}{\bar{\alpha}^t}(\bar{\alpha}^{t+1} - \bar{\alpha}^t).$
 - 9: $t = t + 1.$
 - 10: **end while**
-

After solving the optimization problem (10), the likelihood function is then constructed as

$$p(z_k | x_k^i) \propto \min\{\max\{\alpha_i, 0\}, \max\{\bar{\alpha}_i, 0\}\}, \quad (11)$$

where α_i and $\bar{\alpha}_i$ are the i -entry of the vectors α and $\bar{\alpha}$, respectively. According to this equation the most robust candidate can be selected by using min operator and naturally fusing the information of color and infrared images.

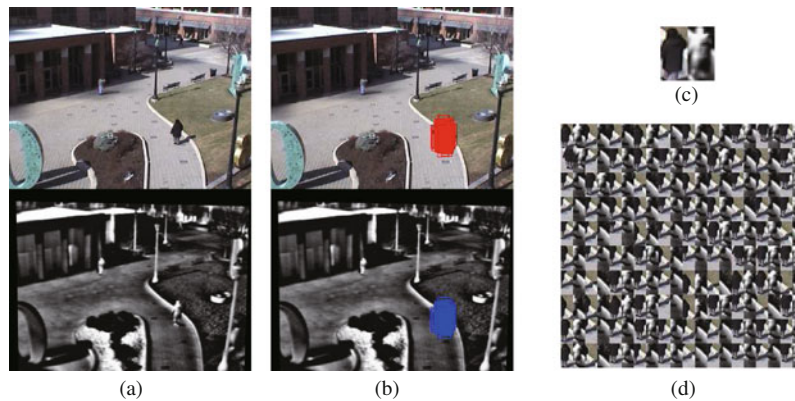


Figure 1 Illustration of particle sampling. (a) The original color and infrared frames; (b) these frames are overlaid by the $N = 100$ boxes which are determined by the particle state x_k^i . Since the color and infrared frames are well registered, they share the same particles; (c) the reference color and infrared templates determined in the first instant, this image is enlarged for better illustration; (d) the particle candidates which include color and infrared samples. Each color and infrared sample corresponding to the particle are resized to 30×15 and are combined to form a sample with size 30×30 .

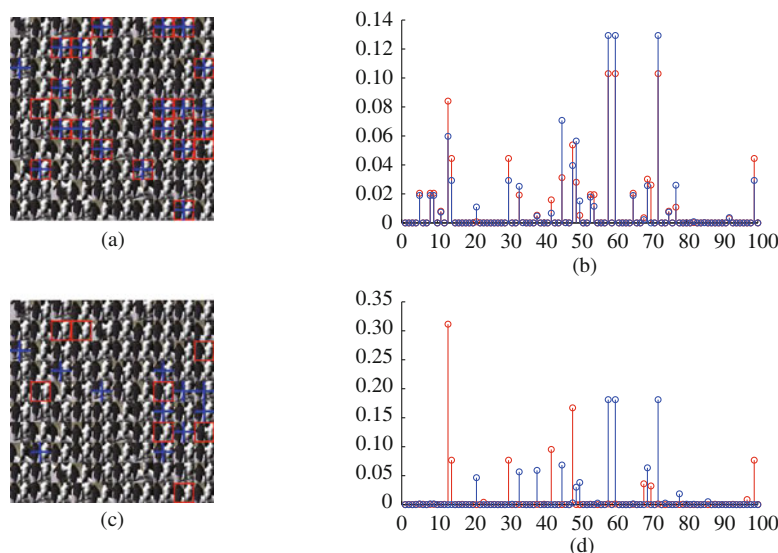


Figure 2 Comparison between joint sparse representation ((a) and (b)) and separate sparse representation ((c) and (d)). (b) and (d) The sparse representation on the $N = 100$ particles, where red line corresponds to the color image and blue line corresponds to the infrared image; (a) and (c) the corresponding particle samples are shown. For each sample, if the corresponding sparse representation coefficient according to color (infrared) feature is larger than 0.01, then it is marked with a red (blue) box.

Remark 1. Prior to solving optimization problem (10), all the feature vectors need to be normalized so that their 2 norms are 1. Normalization is necessary because otherwise the decomposition coefficients would be sensitive to the magnitudes of the feature vectors.

In fusion tracking, some particles may obtain high likelihood in color image, while some particles may obtain high likelihood in infrared image. However, a “good” particle should obtain high likelihood in both color and infrared images. This intuitive observation can be effectively utilized in the proposed joint sparse representation approach. In Figure 2 an illustration of the joint sparse representation is given. This figure shows that the separate sparse representation approach usually selects different particles when using different features, while the joint sparse representation approach is forced to select the same particle when using different feature. In this setting, the constraint of joint sparsity across different information

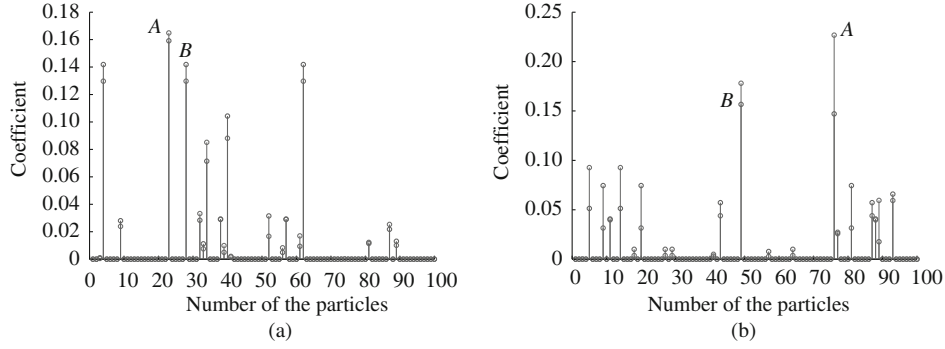


Figure 3 Two cases of using the updating module. (a) The best color particle (see point A) and the best infrared particle (see point B) are the same. (b) the best color particle (see point A) and the best infrared particle (see point B) are different.

sources is valuable since different images may favor different sparse reconstruction coefficients, while the joint sparsity may enforce the robustness in coefficient estimation. This is an obvious advantage when fusing different modal information since joint sparse representation introduces extra robustness[23]. In addition, a computational advantage of this approach is that it requires the joint sparse representation to be computed for only once, while some existing researches require the sparse representation to be computed for each particle[16].

4 Reference models updating using co-learning

In practical tracking scenarios, the reference template $\mathbf{r}_C$ and $\mathbf{r}_I$ should be updated to incorporate the changes of appearance, illumination, and so on. Although a lot of existing update approaches can be used to enhance the tracking performance, most of them belong to “self-learning” style algorithm. That is to say, the reference template determined in the previous frame is used in current frame to evaluate possible regions. Then the weighted sum of the current tracking result and the reference template is used to get new reference template. This class of approaches usually tends to “drift” since the error may be accumulated during the learning and tracking process. To cope with this, some scholars proposed to use semi-supervised learning approach to avoid the drift problem, see, for example, [24–27]. However, all of these approaches are based on classifier-based tracking and cannot be applicable to our case. Ref. [28] proposed a new promising co-learning approach to update the reference template and detector. This idea can be utilized in our framework to design a simple yet effective co-learning approach to update the reference templates in color and infrared images. The details are introduced as follows.

At time instant k , the joint sparse representations α and $\bar{\alpha}$ are given. According to the definition of similarity induced by the joint sparse representation, the sample $\hat{\mathbf{x}}_k^{j_I}$ with $j_I = \operatorname{argmax}_{i=1,2,\dots,N}\{\bar{\alpha}_i\}$ is the most similar to the infrared reference template and the sample $\hat{\mathbf{x}}_k^{j_C}$ with $j_C = \operatorname{argmax}_{i=1,2,\dots,N}\{\alpha_i\}$ is the most similar to the color reference template. To construct the co-learning approach, the infrared feature $\mathbf{t}_I^{j_C}$ can be extracted from sample $\hat{\mathbf{x}}_k^{j_C}$ and the color feature $\mathbf{t}_C^{j_I}$ can be extracted from the infrared sample $\hat{\mathbf{x}}_k^{j_I}$. The references can be updated by the following equation:

$$\begin{cases} \mathbf{r}_C = \mu \mathbf{r}_C + (1 - \mu) \mathbf{t}_C^{j_I}, \\ \mathbf{r}_I = \mu \mathbf{r}_I + (1 - \mu) \mathbf{t}_I^{j_C}, \end{cases} \quad (12)$$

where μ is the updating parameter set by the designer. If μ is set to 1, then the reference templates will keep fixed and therefore μ can be used to control the updating speed. In the presented experiments, the value of μ is set to 0.95.

In practice $j_C = j_I$ occurs often (see points A and B in Figure 3(a)). This means that both color modality and infrared modality select the same “best” particle sample for updating. However, this situation does not hold for all cases (see points A and B in Figure 3(b)). This means that color modality

and infrared modality select different particle samples for mutual updating. In this case, the best color candidate is used to update the infrared reference template and the best infrared candidate is used to update the color reference template. This approach belongs to “co-learning” style and can effectively avoid the template drift problem.

5 Experimental results

The proposed approach was tested on thermal/color video sequence pairs from OTCBVS dataset collection [29]. Data consist of 8-bit gray-scale bitmap thermal images, and 24-bit color bitmap images of 320×240 pixels. All images were sampled approximately at 30 Hz, and were registered using homography with manually-selected points. Thermal sequences were captured using a Raytheon PalmIR 250D sensor, and color sequences were captured using a Sony TRV87 Handycam[6].

In the experimental validation, all algorithms were tested on a 1.86 GHz Pentium PC with 3 GB memory, through MATLAB implementation. For all of the experiments, the states of the particle filter are defined as $\mathbf{x}_k = [x_k, y_k, s_k]^T$, where x_k, y_k indicate the locations of the object in registered color and infrared images, respectively; s_k is the corresponding scale. The dynamics of the tracked object are assumed to be a random walking model, which can be represented as $\mathbf{x}_k = \mathbf{x}_{k-1} + \mathbf{v}_k$, where $\mathbf{v}_k$ is a multivariate zero-mean Gaussian random variable whose variance is set by $[\sigma_x, \sigma_y, \sigma_s] = [2, 2, 0.05]$. The number of all particle filters is set to 100.

In this experiment, the tracking results of three fusion algorithms are compared: separate sparse representation (SSR), covariance tracker (CVT) and the proposed joint sparse representation (JSR) approach.

SSR is a straightforward extension of the approach proposed in [18], serving as the baseline approach. In this approach, the sparse representation in color and infrared images are conducted separately (see Figure 2). The likelihood function defined in [18] is used to design two likelihood functions in color and infrared images, respectively. Finally they are multiplied to form the fused likelihood function. By this comparison the advantages of the proposed joint sparse representation approach can be shown.

Both SSR and JSR need to extract features from color and infrared frames. In color frames, color distributions are used as object models since they achieve robustness against non-rigidity, rotation and partial occlusion. In the experiments, the histograms are typically calculated in the HSV space using $10 \times 10 + 10$ bins. The resulting complete histogram is thus composed of $N_h = 110$ bins. In infrared frames, each sample is resized to 20×10 and appearance is extracted as a 200-dimensional feature vector.

CVT is the approach proposed in [30] and a state-of-the-art tracker which can naturally fuse multiple features or modalities. According to [30], a particle filter approach is formed whose details can be found in [31]. Notice that the feature used in CVT is totally different from that used in SSR and JSR. For covariance construction, except the location, only R, G, B values in color images and intensity value in infrared image are used.

In addition, to show the advantage of the fusion, two non-fusion approaches are compared. One is termed as Color approach, which uses the color image only, and the other is termed as Infrared approach, which uses the infrared image only. Both approaches use the corresponding feature and the likelihood functions with the SSR approach.

In the experimental validation, two sequences are considered. The tracking results on Sequence I (500 frames) is shown in Figure 4. The scenes in this sequence contain a large amount of moving cloud. The Color approach fails when the object enters into the shadow region (see yellow boxes in frames 406 and 500), and the Infrared approach mistakenly locks into the telegraph pole when the object passes it (see cyan boxed in frames 249, 406 and 500). During frames 1–144, the tracking results of the three fusion algorithms are satisfactory since the object does not enter the shadow region. At frame 160, the results of CVT begins to drift and finally locks into another object (see frame 500). The results of SSR begins to deteriorate at frame 406 and never recovers, while JSR persistently tracks the object accurately.

Figure 5 shows selected tracking results on Sequence II with 1411 frames totally. All fusion algorithms are able to track the object from start to frame 1186 but the Color approach and Infrared approach

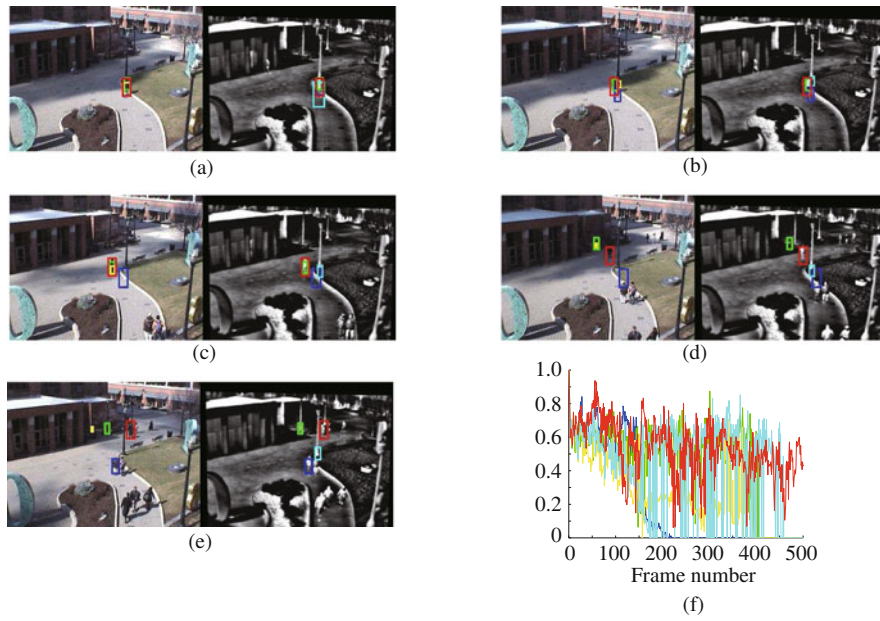


Figure 4 Selected tracking result of Sequence I. (a) Frame 144; (b) frame 160; (c) frame 249; (d) frame 406; (e) frame 500; (f) the tracking accuracy curve (red: JSR approach; green: SSR approach; blue: CVT approach; yellow: Color approach; cyan: Infrared approach).

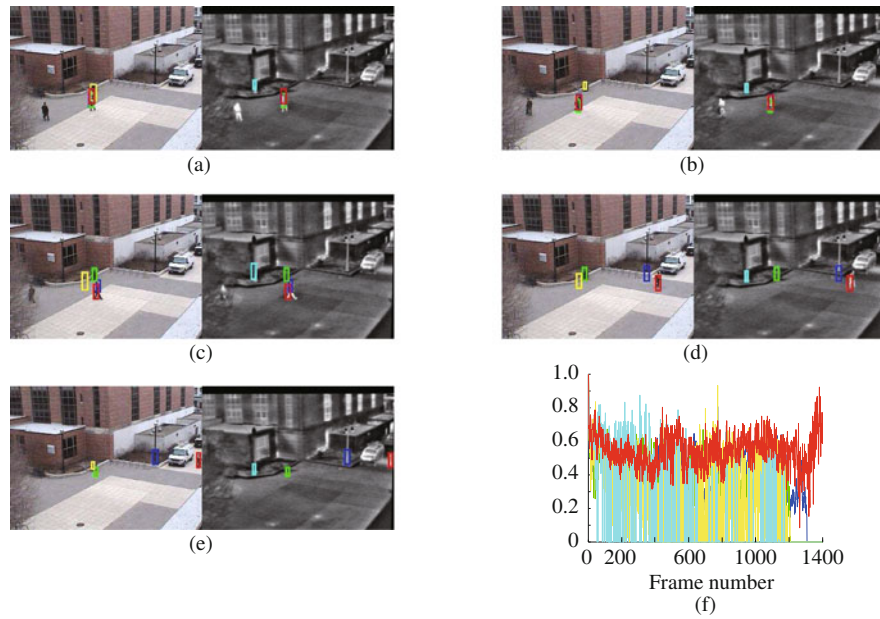


Figure 5 Selected tracking result of Sequence II. (a) Frame 387; (b) frame 1186; (c) frame 1202; (d) frame 1310; (e) frame 1411; (f) the tracking accuracy curve (red: JSR approach; green: SSR approach; blue: CVT approach; yellow: Color approach; cyan: Infrared approach).

lock into mistake objects. Therefore the advantages of the utilization of fusion mechanism can be shown. However, when the object passes the trash at frame 1202, SSR begins to lock the trash and never recovers. At frame 1310, CVT begins to lock the telegraph pole and neve recovers. The proposed JSR, however, is able to track the object till the last frame.

Finally, the ground truth data for the two sequences are manually labelled for quantitative analysis. The tracking accuracy for each frame is evaluated by the overlap ratio, which is defined as the ratio between the area of intersection $GT \cap TR$ and the area of union $GT \cup TR$, where GT and TR are the

Table 1 Tracking performance comparison for Sequence I

Approach	CVT	Color	Infrared	SSR	JSR
Tracking accuracy	0.2068	0.2418	0.4333	0.4379	0.5351
Center error (in pixels)	27.2790	12.3834	9.8797	8.2217	4.1985
Time cost (in seconds)	0.4948	0.8891	0.9899	1.5946	1.2435

Table 2 Parameters for configuration 1

Approach	CVT	Color	Infrared	SSR	JSR
Tracking accuracy	0.4110	0.2369	0.1310	0.3511	0.5417
Center error (in pixels)	7.0428	32.1159	49.9092	20.8460	3.3780
Time cost (in seconds)	0.2042	0.6864	0.6760	1.3085	0.8972

ground truth box and the tracking result box, respectively. The bottom-right panels in Figures 4 and 5 show the overlap ratio of the five algorithms on the two sequences. Tables 1 and 2 give the mean tracking performance comparison, where the tracking accuracy is defined as above, the center error (in pixels) is defined as the Euclidean distance between the centers of the ground truth and the tracking boxes, and the time cost (in seconds) is calculated as the mean processing time through the whole sequence. Both tables show that both SSR and JSR perform better than Color and Infrared which use single modal information source, while JSR provides larger accuracy and smaller error than SSR. CVT needs the smallest time cost since it does not need sparse representation. Both SSR and JSR require more time costs than Color and Infrared since more visual processing and more complicated sparse representations are needed. However, it should be noticed that the time costs of JSR is a little smaller than SSR since SSR requires two times separate sparse representation, while JSR requires one joint sparse representation. In summary, the proposed JSR approach offers a noticeable improvement over both SSR and CVT. In addition, we notice that the tracking performances for Sequence II is better than that for Sequence I. The reason is that in Sequence II, the object moves from the left to right and therefore the depth information changes a little, while in Sequence I, the object moves from near to far and more disturbances are introduced. In sequence II, the object begins to move rapidly and the tracking performance of JSR decreases since we use a single motion model.

6 Conclusions

Joint sparse representation provides a natural and robust approach to fuse different modal information. In this paper, the joint sparse representation is used to design similarity between samples and the likelihood function of particle filter fusion tracker. The proposed approach effectively combines color visual spectrum and thermal spectrum images for object tracking. In addition, a co-learning approach is proposed to update the reference templates of both modalities and enhance the tracking robustness. Finally the proposed fusion scheme is verified using OTCBVS dataset and is shown to outperform state-of-the-art approaches.

Acknowledgements

This work was jointly supported by National Natural Science Foundation of China (Grants No. 61075027, 91120011, 90820304), and National Key Project for Basic Research of China (Grant No. 2007CB311003).

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